

Basel Ahmed Sardah

Data Scientist and AI Specialist engineering end-to-end Machine Learning pipelines and edge-deployed Computer Vision architectures. Proven expertise in translating unstructured datasets into actionable business insights, automating enterprise workflows, and articulating high-dimensional data strategies to non-technical stakeholders to drive operational return on investment.

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WORK EXPERIENCE

Beaee Platform

Software Engineer Intern

Jan/2026 - Jul/2026

- Directed IT workflows and led developer teams, coordinating directly with executive management to align technical data solutions with organizational business goals.
- Engineered the extraction and interpretation of raw platform data into actionable performance reports, presenting comprehensive data-driven visualizations to guide stakeholder evaluation.
- Engineered an automated Python-based parsing pipeline to orchestrate bulk HR scheduling and communication workflows, eliminating manual administrative bottlenecks.
- Collaborated in re-architecting the relational database schema, optimizing data normalization to accelerate querying performance for analytical reporting.
- Containerized server environments and backend API services using Docker, ensuring pipeline consistency and deployment reliability across client applications.

KFU Clubs

Technical Instructor (Algorithms & Math for CS)

2023 - 2026

- Delivered over 30 hours of instructional workshops on Algorithm Analysis and Logic, translating complex computational theories into accessible concepts for non-technical audiences.
- Co-led the editorial team for a technical magazine, authoring over 15 publications to articulate emerging tech trends and complex data insights to broader audiences.

Technical Projects

AI Trust Gateway (IEEE Research) | NLP, BERT, XGBoost, Python

2026

- Engineered a fault-tolerant Natural Language Processing pipeline using a Sliding Window Chunking mechanism to bypass 512-token limits and Out-Of-Memory bottlenecks for high-dimensional data.
- Deployed a low-TCO hybrid classifier combining a frozen bert-base-uncased encoder with XGBoost, achieving 99.64% binary and 85.85% multi-class detection accuracy while minimizing inference latency.
- Outperformed industry baselines by 14%, virtually eliminating false positives to automate fraud-triage systems and significantly reduce costly manual review bottlenecks.

QASSAS (Visual Anomaly Detection Pipeline) | PyTorch, Diffusion Models, OpenCV

2025 - 2026

- Engineered an end-to-end Computer Vision pipeline for micro-defect detection in manufacturing utilizing Diffusion Models and LoRA, achieving an 87% overall accuracy and a 97% Recall rate.
- Formulated rigorous data preprocessing and augmentation logic using Pandas and NumPy to eliminate noise and maintain maximum dataset integrity during model training.

Edge AI Vision System | Python, YOLO, TensorRT, Linux

2026

- Architected an asynchronous edge AI pipeline using NVIDIA TensorRT (FP16), decoupling I/O frame acquisition and inference to sustain 60+ FPS processing without CPU starvation.
- Engineered a deterministic spatial decision engine for hardware actuation via direct Unix socket communication, applying Euclidean hazard avoidance algorithms for automated trajectory mapping.

Gaming Market Penetration Analysis | Python, Power BI, Data Engineering

2025

- Extracted and processed unstructured repository datasets, engineering an ETL pipeline to clean and normalize high-dimensional gaming market data.
- Developed a comprehensive Power BI dashboard to visualize Linux OS market penetration, delivering strategic, data-driven recommendations on platform expansion and ROI potential.

Predictive Analytics Web Dashboard | Python, Scikit-learn, XGBoost, Flutter Web

2025

- Trained and deployed predictive machine learning models using Scikit-learn and XGBoost, achieving a highly precise Mean Absolute Error (MAE) of less than 1.0.
- Built an interactive cross-platform web dashboard allowing non-technical stakeholders to input raw data and visualize complex analytical insights and strategic performance metrics.

Technical Skills

- AI & Machine Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, BERT, YOLO, TensorRT, Diffusion Models.
- Data Science & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Power BI, Tableau, R, BigQuery.
- Programming & Infrastructure: Python, SQL, FastAPI, Docker, Git/GitHub, Linux (Arch/Wayland).

EDUCATION and Certifications

- King Faisal University Bachelor of Computer Science with GPA 4.79 / 5.00 (2021 - 2026) .
- AI Machine Learning Specialization from DeepLearningAI (2025).
- Google Data Analytics Professional Certificate (2024).
- University of Michigan Intro to Data Science in Python (2025).